SC-MCU-100 Series

32-bit Automotive Microcontroller for Vehicle Computers and Zone Controllers

Datasheet — Revision 1.2

Fictional document. The SC-MCU-100 is a fictional device invented for demonstrating the SpecCheck datasheet verification tool. Any resemblance to real products is coincidental. Do not use for design decisions.

1 Key features

- Dual-core lockstep 32-bit RISC CPU with floating point unit
- 4096 KB embedded program flash with ECC (SEC-DED)
- Up to 1 MB SRAM with ECC, depending on device variant (see Table 7-1)
- 3 CAN FD controllers compliant with ISO 11898-1:2015
- 1 Ethernet MAC (10/100 Mbit/s) with RMII interface for 100BASE-T1 PHYs
- Hardware security module with AES-128/256 and true random number generator
- Functional safety: designed for ISO 26262 ASIL-B applications
- Packages: LQFP-144 and BGA-292

2 Device overview

The SC-MCU-100 integrates two CPU cores in lockstep, a crossbar interconnect, embedded memories and a rich set of automotive communication peripherals. The device targets body control, gateway and zone-controller functions inside a vehicle computer architecture.

2.1 CPU subsystem

The CPU subsystem consists of a primary core and a checker core executing the same instruction stream with a two-cycle delay. Mismatches are reported to the fault collection unit. The maximum CPU frequency is specified in the electrical characteristics (Table 5-1).

2.2 Memory subsystem

Program flash: 4096 KB organised in four banks of 1024 KB, read-while-write capable. All flash and SRAM arrays are protected by error correction code (ECC) with single-error correction and double-error detection.

The SRAM size depends on the device variant. Refer to the ordering information in Section 7 for the memory configuration of each part number.

3 Communication interfaces

Interface	Instances	Notes
CAN FD	3	MCAN0 – MCAN2, up to 8 Mbit/s data phase
LIN	4	LIN 2.2A / SAE J2602
Ethernet MAC	1	10/100 Mbit/s, RMII, IEEE 802.1AS time sync
SPI	6	Up to 50 MHz
I2C	2	Standard, fast and fast-plus mode

Table 3-1: Communication interface overview

3.1 CAN FD

The device contains 3 CAN FD controllers (MCAN0 to MCAN2). Each controller provides 64 message buffers and supports partial networking with an external transceiver.

3.2 Ethernet

The Ethernet MAC connects to an external automotive Ethernet PHY such as a 100BASE-T1 transceiver through the RMII interface.

4 Functional safety and quality

The SC-MCU-100 was developed according to an ISO 26262 compliant process and offers safety mechanisms including lockstep cores, memory ECC, clock monitors and a windowed watchdog. A safety manual is available on request.

Automotive grade devices are available for applications requiring extended reliability.[1] Contact your sales representative for production part approval documentation.

The hardware security module provides secure boot, key storage and cryptographic acceleration for secure on-board communication.

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[1] Devices with the suffix "-Q" are qualified according to AEC-Q100 Grade 1.

5 Electrical characteristics

5.1 Recommended operating conditions

Symbol	Parameter	Min	Typ	Max	Unit
TA	Ambient operating temperature	-40	-	125	°C
TJ	Junction temperature	-40	-	150	°C
VDD	Core supply voltage	1.14	1.20	1.26	V
VDDIO	I/O supply voltage	2700	3300	5500	mV
fCPU	CPU clock frequency	-	-	400	MHz

Table 5-1: Recommended operating conditions

5.2 Supply current

Mode	Typ	Max	Unit
Run, 400 MHz, all peripherals	180	260	mA
Standby, SRAM retention	45	120	µA

Table 5-2: Supply current

6 Package information

Package	Body size	Pitch	GPIO
LQFP-144	20 × 20 mm	0.5 mm	112
BGA-292	17 × 17 mm	0.8 mm	220

Table 6-1: Package options

Thermal resistance values are provided in the separate package application note.

7 Ordering information

Part number	Flash	SRAM	Package	Grade
SC-MCU-100A	4096 KB	512 KB	LQFP-144	Industrial
SC-MCU-100B	4096 KB	1024 KB	BGA-292	Industrial
SC-MCU-100B-Q	4096 KB	1024 KB	BGA-292	Automotive

Table 7-1: Ordering information

Not all combinations are available in all regions. SRAM size of engineering samples may differ from production devices.

8 Revision history

Revision	Date	Changes
1.0	2025-03-01	Initial release
1.1	2025-07-15	Added BGA-292 package
1.2	2025-11-20	Updated ordering information

This datasheet is fictional and was generated by a script in the SpecCheck repository. It contains deliberate edge cases for testing.